Unit 1

Syllabus:
Network Hardware: Local Area Networks, Metropolitan Area Networks, Wide Area Networks, Internetwork. **Network software:** Protocol Hierarchies, Design Issues for the Layers, Connection-Oriented and Connectionless Services. **Reference Models:** OSI, TCP/IP Reference models, Comparison of OSI and TCP/IP Models **Example Networks**: ARPANET, Internet. **Physical Layer:** Guided Transmission media: Magnetic Media, Twisted pairs, Coaxial Cable, Fiber Optics, Unguided Transmission Media: Radio waves, Microwaves, Infrared.

Definition : Computer network is a system of two or more interconnected computing devices that communicate with each other to share data, resources, and services. These devices often called **end devices(also referred to as an host)** can include anything from traditional desktop computers and servers to smartphones, smart TVs, and IoT (Internet of Things) sensors. They connect through **physical wires** (like fiber-optic or ethernet cables) or **wireless technologies** (like Wi-Fi, 5G, and Bluetooth).

Key Components of a Network

1. **Host (End Devices):** Any device that can send, receive, or process data (e.g., PCs, laptops, printers, routers).
2. **Transmission Media:** The physical or wireless pathway data travels through (e.g., twisted-pair cables, fiber optics, radio waves).
3. **Protocols:** The standardized rules that dictate how devices package, send, and interpret data (e.g., TCP/IP, HTTP, DNS).
4. **Network Devices:** Specialized hardware that directs traffic across the network (e.g., switches, routers, firewalls).

Primary Benefits

1. **Resource Sharing**: Multiple devices can share physical hardware (like printers) or software licenses, cutting costs.
2. **Data Sharing:** Allows seamless file exchange and access to centralized databases.
3. **Communication:** Enables instant messaging, video calls, email, and collaborative tools across distances.
4. **Centralized Security & Management**: Admins can manage permissions, software updates, and security protocols for hundreds of devices from a single point.

Key Functions of an End Device

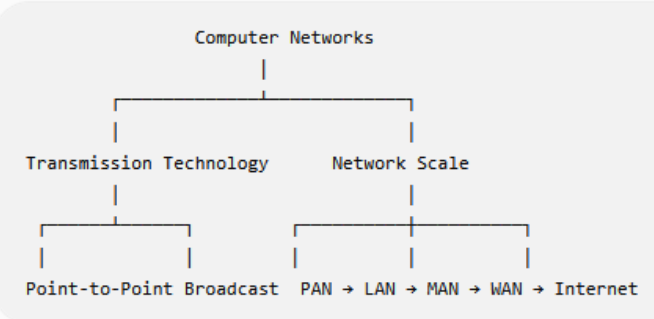
1. **Interface with Users**: End devices provide the screen, keyboard, or sensors needed to interact with humans or physical environments.
2. **Initiate Communication**: When you open a website, your laptop (end device) creates a request packet and sends it onto the network.
3. **Terminate Communication**: When a remote server sends back the web page data, your laptop receives, decodes, and renders it on screen.

key functions of intermediate devices:

1. The primary job of intermediate devices is to determine the best path for data to travel across a network.
2. Signal Restoration: Devices like repeaters, hubs, and switches receive weak or distorted signals, clean them up, and regenerate them at full strength before retransmitting them.
3. Extending Range: This allows network connections to span across long physical distances without

Network Hardware: Local Area Networks, Metropolitan Area Networks, Wide Area Networks, Internetwork.

Network Classification : Computer networks are classified based on transmission technology and scale.



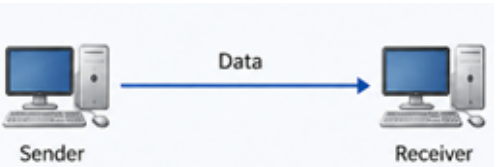
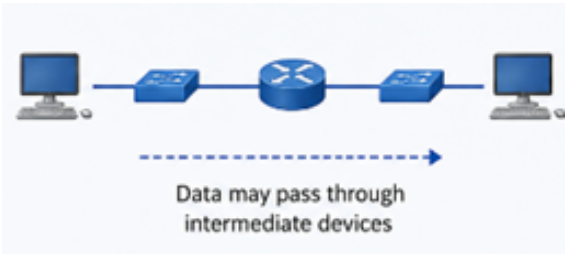
Transmission Technology

Transmission technology refers to the method used to transfer data between devices in a computer network. Based on the communication method, networks are classified into two types:

1. Point-to-Point Links
2. Broadcast Links

1. Point-to-Point Links

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Aspect	Description
Definition	A Point-to-Point (P2P) link is a dedicated communication link established between exactly two devices. Data is transmitted directly from the sender to the receiver.
Characteristics	• Connects only two devices. • Uses a dedicated communication path. • Data reaches only the intended receiver. • Provides higher security and privacy. • Easier to detect and troubleshoot faults. • Multiple links are required to connect many devices.
Advantages	• High speed due to dedicated bandwidth. • Secure communication. • Reliable data transmission. • Less interference from other devices.
Disadvantages	• Expensive for large networks. • Requires many cables if many devices are connected.
Applications	• Expensive for large networks. • Requires many cables if many devices are connected.
Diagram	 or 

Broadcast Links

A **Broadcast link** is a communication method where **one device transmits data to all devices** connected to the same shared communication medium. Every device receives the transmission, but only the intended recipient processes the data.

Aspect	Description
Definition	A **Broadcast link** is a communication method in which a single sender transmits data over a **shared communication channel**, allowing all connected devices to receive the transmission. Only the device with the matching destination address processes the data.
Characteristics	• One sender communicates with multiple receivers.• Uses a shared communication channel.• Every device receives the transmitted frame.• Destination address determines which device accepts the data.• Suitable for small and medium-sized local networks.
Advantages	• Less cabling required.• Easy to add new devices.• Cost-effective.• Efficient for sending common information to multiple devices.
Disadvantages	• Lower security.• Shared bandwidth may cause congestion.• Collisions can occur .• Performance decreases as the number of devices increases.
Applications	• Traditional Ethernet LANs using hubs.• Wi-Fi (Wireless LANs).• Television broadcasting.• Radio broadcasting.• Satellite communication.
Diagram	

Comparison of Point-to-Point and Broadcast Links

Feature	Point-to-Point	Broadcast
Number of devices	Two devices	Multiple devices
Communication path	Dedicated link	Shared communication medium
Data delivery	Only intended receiver	All devices receive the transmission
Bandwidth	Dedicated	Shared
Security	High	Lower
Cost	Higher	Lower
Performance	Better	May decrease with more users
Examples	Router-to-router, leased line, telephone	Ethernet LAN, Wi-Fi, TV broadcasting

1. Protocol Hierarchies,
2. Design Issues for the Layers,
3. Connection-Oriented and Connectionless Services.

Protocol Hierarchies

A **protocol hierarchy** is a structured arrangement of communication protocols organized into multiple layers, where each layer performs a specific function in the communication process and provides services to the layer above it while using the services of the layer below it. To reduce their design complexity, most networks are organized as a stack of layers or levels, each one built upon the one below it.

1. **Interface** : Between each pair of adjacent layers is an **interface**. The interface defines which primitive operations and services the lower layer makes available to the upper one.
2. **Layers:** The purpose of each layer is to offer certain services to the higher layers while shielding those layers from the details of how the offered services are actually implemented. The number of layers, the name of each layer, the contents of each layer, and the function of each layer differ from network to network.
3. **Layer n protocol** : Layer n on one machine carries on a conversation with layer n on another machine, the rules and conventions used in this conversation are collectively known as the layer n protocol.
4. **Protocols:** protocol is an agreement between the communicating parties on how communication is to proceed
5. **Network architecture** : A set of layers and protocols is called a network architecture
6. **Protocol stack :** list of the protocols used by a certain system, one protocol per layer, is called a protocol stack.
7. **Peers** : The entities comprising the corresponding layers on different machines are called peers.

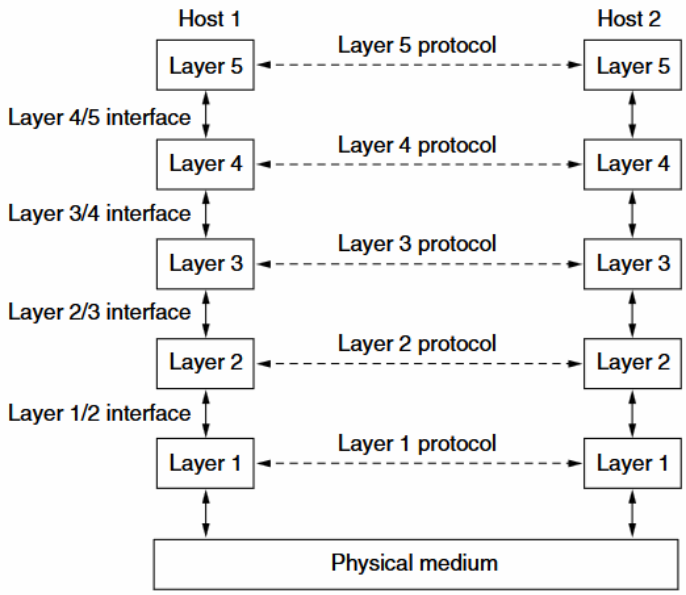
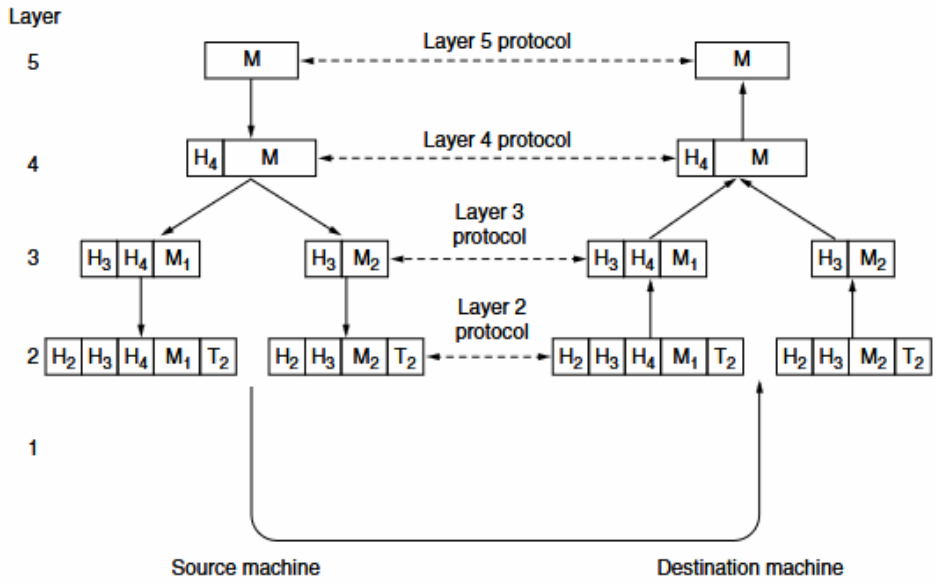


Figure 1-13. Layers, protocols, and interfaces.

Communication Through the Five-Layer Network architecture



1. An application at the Application Layer (Layer 5) creates a message (M) and passes it to the Transport Layer (Layer 4).
2. The Transport Layer adds a header containing control information (such as addresses, sequence numbers, message size, and timing information) and forwards the message to the Network Layer (Layer 3).
3. If the message is larger than the maximum packet size supported by the Network Layer, it is divided into smaller packets (e.g., M and M), and each packet receives its own Network Layer header.
4. The Data Link Layer (Layer 2) adds a header and trailer to each packet, creating frames, and sends them to the Physical Layer (Layer 1).
5. The Physical Layer transmits the data as bits over the communication medium.
6. At the receiving computer, the data moves up the protocol stack, where each corresponding layer removes its own header (and trailer) and passes the remaining data to the next higher layer until the original message reaches the destination application.
7. Communication between corresponding layers on different devices is logical (virtual) and follows a protocol, while the actual data is transmitted through the lower layers and their interfaces.

Design Issues for the Layers,

Computer network design focuses on ensuring reliable communication, scalability, efficient resource utilization, traffic management, interoperability, and secure data transmission

Key Design Issues in Computer Networks

1. Reliability : The network should deliver data **correctly and without errors**. If data is damaged or lost, it should be **detected and sent again**.

Example: If a file is corrupted during download, the missing data is retransmitted.

2. Routing : The network should choose the **best path** for data to travel. If one path fails, it should automatically select another path.

Example: If one road is blocked, GPS finds an alternate route.

3. Addressing : Every device on the network must have a **unique address**. This helps send data to the correct destination.
Example: An IP address is like a house address for a computer.

4. Internetworking: Different types of networks should be able to **communicate with each other**. Large messages may be divided into smaller packets and reassembled at the destination.
Example: The Internet connects Wi-Fi, mobile networks, and wired networks.

5. Scalability : The network should work efficiently even when the number of users or devices increases.
Example: A university network should still perform well when thousands of students are connected.

6. Resource Allocation : Network resources such as **bandwidth** should be shared fairly among users.
Example: Multiple students can use the college Wi-Fi at the same time.

7. Flow Control : A fast sender should not send data faster than a slow receiver can handle.
Example: A teacher speaks slowly if students cannot take notes quickly.

8.Congestion Control : **Congestion** occurs when the network becomes overloaded with excessive data traffic. A common strategy to control congestion is for each computer to reduce its data transmission rate whenever it detects network congestion.

9. Quality of Service (QoS) : Important applications should receive better network service. Real-time applications need faster delivery than normal data.
Example: During a video call, voice and video are given higher priority than file downloads.

10. Security : The network should protect data from unauthorized access or modification. It provides:
Confidentiality: Keeps data private.,**Authentication:** Confirms the user's identity ,**Integrity:** Ensures data is not changed during transmission.

Example: Online banking uses encryption to keep your information secure

Connection-Oriented and Connectionless Services.

Network services define how data is transmitted between communicating devices. The two main types of network services are **Connection-Oriented Service** and **Connectionless Service**. They differ in how connections are established, data is transmitted, and reliability is ensured.

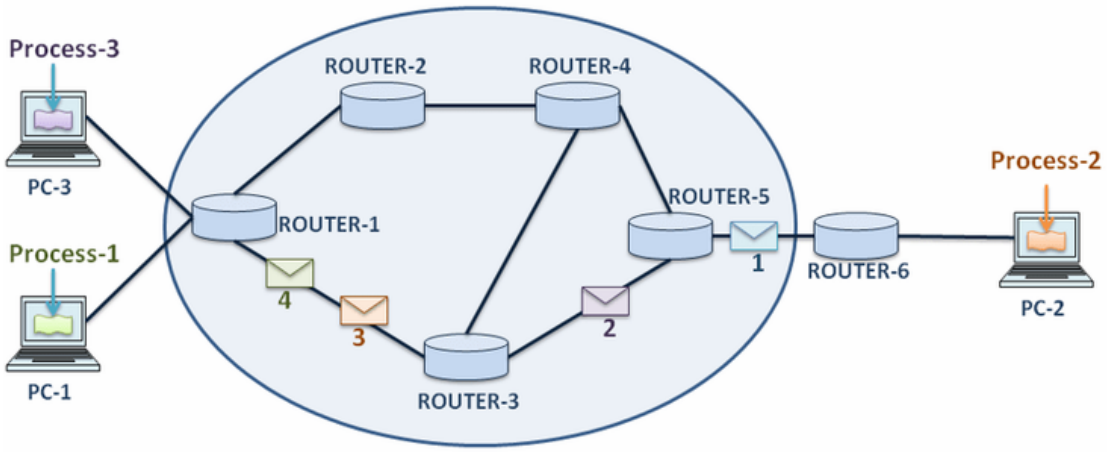
1. Connection-Oriented Service

Definition : A **Connection-Oriented Service** is a communication method in which a logical connection is established between the sender and receiver before data transmission begins. The connection is maintained during communication and is terminated after all data has been transferred.

Working Process

1. **Connection Establishment** – A connection is created between the sender and receiver.
2. **Data Transfer** – Data is transmitted in the correct sequence through the established connection.
3. **Connection Termination** – The connection is closed after communication is complete.

Aspect	Description
Characteristics	Requires connection establishment before communication.Ensures reliable data delivery.Data packets arrive in the correct order.Provides error detection, acknowledgments, and retransmission.Suitable for long-duration communication.
Advantages	Reliable communication.Ordered delivery of packets.Error recovery through retransmission.Flow and congestion control improve network performance.
Disadvantages	Higher overhead due to connection setup.More delay before data transmission.Requires more memory and processing.
Applications	File Transfer (FTP).Email (SMTP).Web browsing (HTTP over TCP).Online banking.Video conferencing.



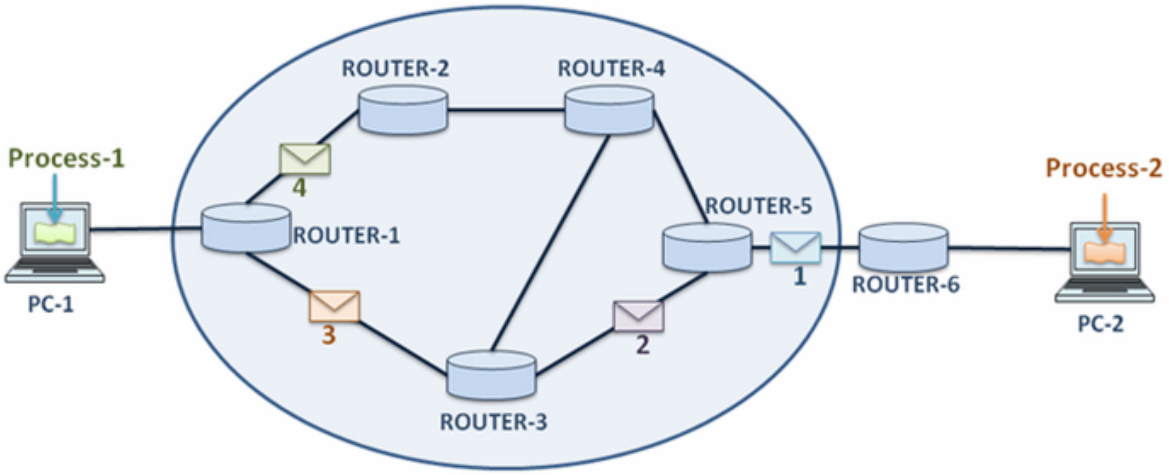
Connectionless Service

Definition : A **Connectionless Service** is a communication method in which data is transmitted without establishing a dedicated connection between the sender and receiver. Each packet is treated independently and may follow different paths to reach the destination**.**

Working Process

1. Sender directly transmits packets.
2. Each packet is routed independently.
3. Receiver processes packets as they arrive.

Aspect	Description
Characteristics	• No connection establishment required.• Packets travel independently.• Packets may arrive out of order.• No guarantee of delivery.• Faster transmission with low overhead.
Advantages	• Faster communication.• Low transmission overhead.• Efficient for real-time applications.• Better utilization of network resources.
Disadvantages	• No guarantee of packet delivery.• Packets may be lost or duplicated.• Packets may arrive out of sequence.• No built-in error recovery.
Applications	• Live video streaming.• Voice over IP (VoIP).• Online gaming.• DNS (Domain Name System).• Multimedia broadcasting.



Comparison Between Connection-Oriented and Connectionless Services

Feature	Connection-Oriented Service	Connectionless Service
Connection	Required before communication	Not required
Reliability	High	Low
Packet Delivery	Guaranteed	Not guaranteed
Packet Order	Maintained	May not be maintained
Error Recovery	Supported	Not supported
Acknowledgment	Required	Not required
Speed	Slower	Faster
Overhead	High	Low
Suitable For	File transfer, email, banking	Streaming, VoIP, gaming
Example Protocol	TCP	UDP